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# POPULATION ANALYSIS OF LOTKA-VOLTERRA PREDATOR-PREY DYNAMICS: A NUMERICAL SIMULATION OF COUPLED POPULATION RESPONSE UNDER IDEALIZED ECOLOGICAL MODELING ASSUMPTIONS

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## NOMENCLATURE

| | |
|---|---|
| $t$ | time, years |
| $x$ | prey population |
| $y$ | predator population |
| $\frac{dx}{dt}$ | rate of change of prey population |
| $\frac{dy}{dt}$ | rate of change of predator population |
| $\alpha$ | prey growth rate constant, 1 |
| $\beta$ | predation interaction coefficient, 0.02 |
| $\gamma$ | predator death rate constant, 1 |
| $\delta$ | predator reproduction coefficient, 0.01 |
| $xy$ | predator-prey interaction term |
| $Y$ | population vector |
| $Y_i$ | initial population vector |
| $Tspan$ | simulation time span |
| *ROC* | rate of change vector |
| *ode45* | MATLAB ordinary differential equation solver |
| *PredPrey* | user-defined MATLAB function for the predator-prey model |

## 1. INTRODUCTION

This project presents a numerical simulation of the Lotka-Volterra predator-prey model to study the time-dependent interaction between two coupled biological populations. Using MATLAB and the ode45 solver, the governing first-order nonlinear differential equations were evaluated over a 35-year period with specified initial conditions and baseline parameter values for prey growth $\alpha$, predation rate $\beta$, predator death $\gamma$, and predator reproduction $\delta$. In addition to the baseline case, a sensitivity analysis was performed by individually doubling each model parameter to examine how changes in system coefficients affect the population response. The resulting plots and outputs provide a clear representation of oscillatory predator-prey behavior and demonstrate how strongly the system dynamics depend on the selected model parameters.

## 2. MATERIALS AND METHODS

### 2.1 Materials

- MATLAB software for numerical computation and simulation
- A personal computer for code development and execution
- Lotka-Volterra predator-prey differential equations
- Baseline model parameters: $\alpha$ =1, $\beta$ = 0.02, $\gamma$ =1, $\delta$=0.01
- Initial population conditions of 30 prey and 15 predators
- Simulation time interval of 35 years
- *Mathematical Biology* Textbook(online) to use as reference for methods used

### 2.2 Lotka-Volterra Predator-Prey Model

The predator-prey system was modeled using the Lotka-Volterra equations, which describe the coupled interaction between predator and prey populations over time. The model used the first-order governing equations:

Equation 2.21 $$\frac{dx}{dt} = \alpha x - \beta xy \quad [1]$$

Equation 2.22 $$\frac{dy}{dt} = -\gamma y + \delta xy \quad [1]$$

where $x$ represents the prey population, $y$ represents the predator population, and t represents time. The parameters $\alpha$ represents the growth rate of the prey population, $\beta$ represents the predation rate, $\gamma$ represents the death rate of the predator population, and $\delta$ represents predator reproduction rate. For this experiment, we set the baseline parameters to $\alpha = 1$, $\beta = 0.02$, $\gamma = 1$, and $\delta = 0.01$.

### 2.3 Integration with an Initial Value

Because the governing equations give us the rate of change, we can obtain the population functions by integrating with respect to time and using the initial conditions that there are 30 prey and 15 predators. For a general variable $x$:

Equation 2.31 $$x(t) = x_0 + \int_{t_0}^{t} \frac{dx}{dt}\, dt \quad [2]$$

where $x(t)$ is a function of $t$, $x_0$ is the initial value when $t$ is 0, $\frac{dx}{dt}$ is the change of $x$ with respect to time, $t_0$ is the initial time (usually 0), and $t$ being time. The same idea can be extended to a coupled system of equations by writing the dependent variables together in vector form. For our model, the population vector is written as:

Equation 2.32 $$Y(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

where $x(t)$ is the prey population and $y(t)$ is the predator population. The initial-value population vector is therefore:

Equation 2.33 $$Y_i = \begin{bmatrix} 30 \\ 15 \end{bmatrix}$$

Applying the same initial-value integration idea as before with $x$, we can express the coupled system as:

Equation 2.34 $$Y(t) = Y_i + \int_{t_0}^{t} \frac{dY}{dt}\, dt$$

where $\frac{dY}{dt}$ represents the rate-of-change vector (ROC) of the system. This shows that the population vector at any later time is equal to the initial population plus the accumulated change in both populations over the interval of interest.

## 2.4 Application to the Lotka-Volterra System

For this study, the rate-of-change vector $\frac{dY}{dt}$ is defined by the Lotka-Volterra equations [eqn 2.21][eqn 2.22]. Therefore, the coupled system can be written as:

Equation 2.41 $$\frac{dY}{dt} = \begin{bmatrix} \alpha x - \beta xy \\ -\gamma y + \delta xy \end{bmatrix}$$

Putting this ROC vector into the initial value equation [eqn 2.34] gives the final equation:

Equation 2.42 $$Y(t) = Y_i + \int_{t_0}^{t} \begin{bmatrix} \alpha x - \beta xy \\ -\gamma y + \delta xy \end{bmatrix} dt$$

This equation allows us to track the predator and prey populations at any time *t*.

## 2.5 MATLAB Implementation

### 2.51 Establish Initial Conditions and Model Parameters

```
%% MODEL PARAMETERS
Tspan = 0:.01:35;
Yi = [30;15];
alpha = [1,2,1,1,1];
beta = [0.02,0.02,0.04,0.02,0.02];
gamma = [1,1,1,2,1];
delta = [0.01,0.01,0.01,0.01,0.02];
```

This section establishes the simulation time span, the initial population conditions, and the parameter values required for the predator-prey model. The time interval is defined as Tspan = 0:0.01:35, which sets the simulation to run from 0 to 35 years in increments of 0.01 years. The initial population vector is defined as Yi = [30;15], where the first half represents the initial prey population and the second half represents the initial predator population [eqn 2.33]. We then created arrays for the four Lotka-Volterra parameters, $\alpha$, $\beta$, $\gamma$, and $\delta$, so that multiple simulations can be ran and analyzed. The first entry of each array corresponds to the baseline case, while the remaining entries are arranged so that one parameter at a time is doubled so that we can conduct sensitivity analysis.

### 2.52 Defining the Rate-Of-Change Function

```
%% DEFINING RATE OF CHANGE FUNCTION
function ROC = PredPrey(~,Y,alpha,beta,gamma,delta)
    x = Y(1);
    y = Y(2);
    PreyROC = alpha*x - beta*x*y;
    PredROC = (gamma*(-1))*y + delta*x*y;
    ROC = [PreyROC;PredROC];
end
```

Then, we defined the ROC function required to evaluate the Lotka-Volterra system at each point in time. The function is written as PredPrey(~,Y,alpha,beta,gamma,delta), where Y is the current population vector and the remaining inputs are the Lotka-Volterra parameters. Inside the function, the first entry of Y is assigned to x and the second entry of the function being assigned to y so that the two populations can be worked with separately. Then, we established the ROC equations [eqn 2.21][eqn 2.22] and input them into a ROC vector, with the prey rate of change PreyROC going into the first entry and the predator rate of change PredROC going into the second entry, similar to what we see happening in equation 2.41.

### 2.53 Solving the System With ode45

```
%% INTEGRATING WITH ODE45 FOR POPULATION
[t, Y]  =
ode45(@(t,Y)PredPrey(t,Y,alpha(1),beta(1),gamma(1),delta(1
)),Tspan,Yi);
[ta,Ya] =
ode45(@(t,Y)PredPrey(t,Y,alpha(2),beta(2),gamma(2),delta(2
)),Tspan,Yi);
[tb,Yb] =
ode45(@(t,Y)PredPrey(t,Y,alpha(3),beta(3),gamma(3),delta(3
)),Tspan,Yi);
[tg,Yg] =
ode45(@(t,Y)PredPrey(t,Y,alpha(4),beta(4),gamma(4),delta(4
)),Tspan,Yi);
[td,Yd] =
ode45(@(t,Y)PredPrey(t,Y,alpha(5),beta(5),gamma(5),delta(5
)),Tspan,Yi);
```

This section uses the MATLAB solver ode45 to numerically solve the coupled predator-prey system over the full interval Tspan. The solver is called once for the baseline case and four more times for the sensitivity-analysis cases, where one parameter at a time is doubled. In each solver call, PredPrey is used as the function that returns the ROC vector. Tspan provides the time interval of 0-35 years with increments of 0.01 years, and Yi supplies the initial population vector of 30 prey and 15 predators. The first solution, [t,Y], corresponds to the baseline case, while [ta,Ya], [tb,Yb], [tg,Yg], and [td,Yd] correspond to the cases where $\alpha$, $\beta$, $\gamma$, and $\delta$ are doubled, respectively.

### 2.54 Generating Population Plots

```
%% GENERATING POPULATION PLOTS
Ypred = [Y(:,2),Ya(:,2),Yb(:,2),Yg(:,2),Yd(:,2)];
Yprey = [Y(:,1),Ya(:,1),Yb(:,1),Yg(:,1),Yd(:,1)];
```

First, we grouped the predator populations from all five simulations into the matrix Ypred. Then, we did the same thing for the prey populations, taking from all five simulations and putting it into the matrix Yprey. This makes it possible to plot all cases at once for comparison.

```
figure(1)
plot(t,Y)
title('Time(years) vs. Population(Baseline)')
xlabel('Time(years)')
ylabel('Population')
legend('PreyPop','PredPop')

figure(2)
plot(t,Yprey)
title('Time(years) vs. Prey(All)')
legend('Baseline','Alpha x2','Beta x2','Gamma x2','Delta
x2')
xlabel('Time(years)')
ylabel('Prey Population')

figure(3)
plot(t,Ypred)
title('Time(years) vs. Predator(All)')
legend('Baseline','Alpha x2','Beta x2','Gamma x2','Delta
x2')
xlabel('Time(years)')
ylabel('Predator Population')

figure(4)
```

```
plot(Yprey,Ypred)
title('Prey Populations vs. Predator Populations')
legend('Baseline','Alpha x2','Beta x2','Gamma x2','Delta
x2')
xlabel('Prey Population')
ylabel('Predator Population')
```

figure(1) plots the baseline predator and prey population together on the same graph to show how the two populations vary over the interval of 35 years. figure(2) plots prey population versus time for the baseline case and the other cases where a singular model parameter is doubled. figure(3) does the same for predator population vs time. figure(4) plots predator population against prey population for all cases, ultimately creating a phase-style plot that shows the relationship between the two populations as they change together. Titles, axis labels, and legends are included in each figure so that the data can be clearly identified and compared.

### 2.55 Tabulating Data

```
%% TABULATING DATA
i=1:100:length(t);
TimeYears = [t(i,1)];
PredPop = [Y(i,2)];
PreyPop = [Y(i,1)];
tbl = table(TimeYears,PredPop,PreyPop);
disp(tbl)
```

This section extracts selected values from the baseline [t,Y] and organizes them into a table. The line i=1:100:length(t); creates a matrix that selects every 100th point from the time array, ultimately reducing the total number of displayed values while still being able to demonstrate the overall population trend. The code takes the corresponding time values from t and stores them in TimeYears. Then, the predator population values are stored in PredPop and the prey population values are stored in PreyPop. These three arrays are then combined into the table tbl = table(TimeYears, PredPop, PreyPop), which arranges the selected results into 3 columns: time, predator population, and prey population. Finally, disp(tbl) displays the table into the command window so data can be viewed.

## 3. DATA

### 3.1 Parameters and Results

Initial Predator Population: 15 predators

Initial Prey Population: 30 preys

Baseline Prey Growth Rate, α: 1

Baseline Predation Coefficient, β: 0.02

Baseline Predator Death Rate, γ: 1

Baseline Predator Reproduction Coefficient, δ: 0.01

Simulation Time Interval: 0 to 35 years

Time Increment: 0.01 years

Sensitivity Cases: α×2, β×2, γ×2, δ×2

### 3.2 Figures, Models, and Tables

| Table 3.21 | | | | | |
|---|---|---|---|---|---|
| Time (years) | Predators | Preys | Time (years) | Predators | Preys |
| 0 | 15 | 30 | 18 | 48 | 318 |
| 1 | 9 | 65 | 19 | 158 | 86 |
| 2 | 9 | 150 | 20 | 86 | 19 |
| 3 | 31 | 300 | 21 | 37 | 16 |
| 4 | 154 | 136 | 22 | 17 | 26 |
| 5 | 102 | 23 | 23 | 9 | 56 |
| 6 | 44 | 15 | 24 | 8 | 129 |
| 7 | 20 | 23 | 25 | 21 | 276 |
| 8 | 10 | 47 | 26 | 137 | 197 |
| 9 | 8 | 109 | 27 | 118 | 29 |
| 10 | 15 | 242 | 28 | 52 | 15 |
| 11 | 106 | 262 | 29 | 23 | 20 |
| 12 | 134 | 39 | 30 | 11 | 40 |
| 13 | 62 | 16 | 31 | 8 | 91 |
| 14 | 27 | 19 | 32 | 12 | 208 |
| 15 | 13 | 35 | 33 | 72 | 307 |
| 16 | 8 | 78 | 34 | 150 | 58 |
| 17 | 10 | 179 | 35 | 74 | 17 |

**TABLE 3.21:** BASELINE PREDATOR AND PREY POPULATIONS OVER TIME

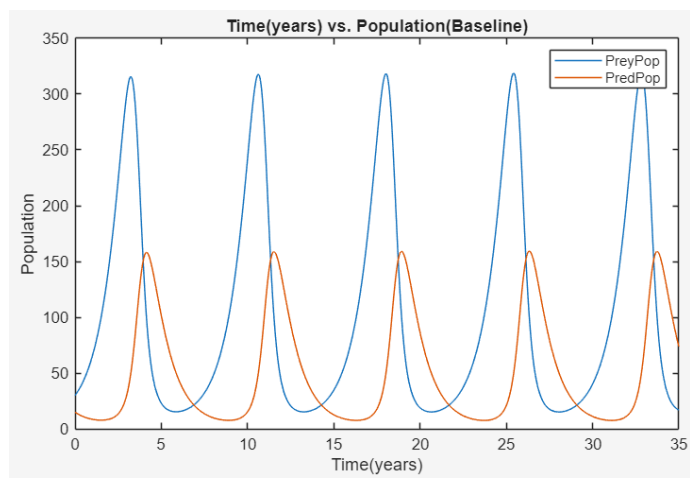


**FIGURE 3.22:** BASELINE PREDATOR AND PREY POPULATIONS OVER TIME

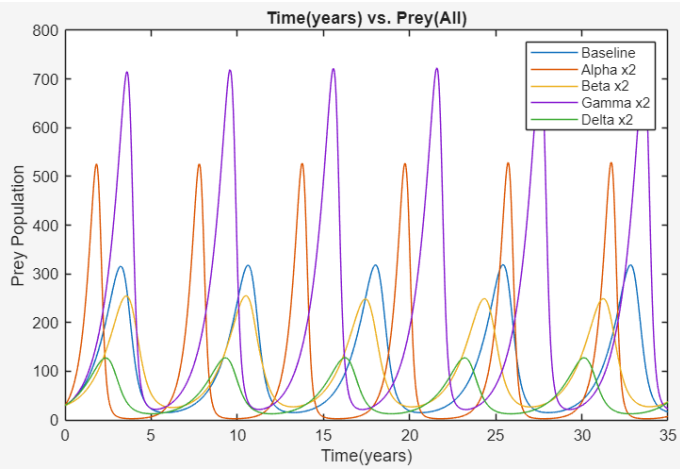


**FIGURE 3.23:** BASELINE, α×2, β×2, γ×2, AND δ×2 PREY POPULATIONS OVER TIME

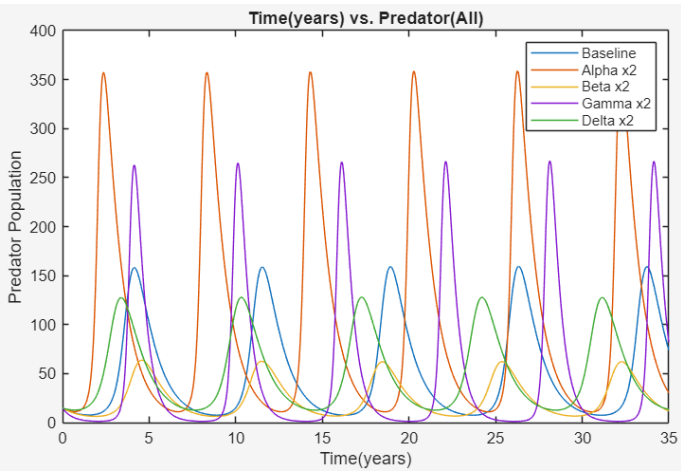


**FIGURE 3.24:** BASELINE, α×2, β×2, γ×2, AND δ×2 PREDATOR POPULATIONS OVER TIME

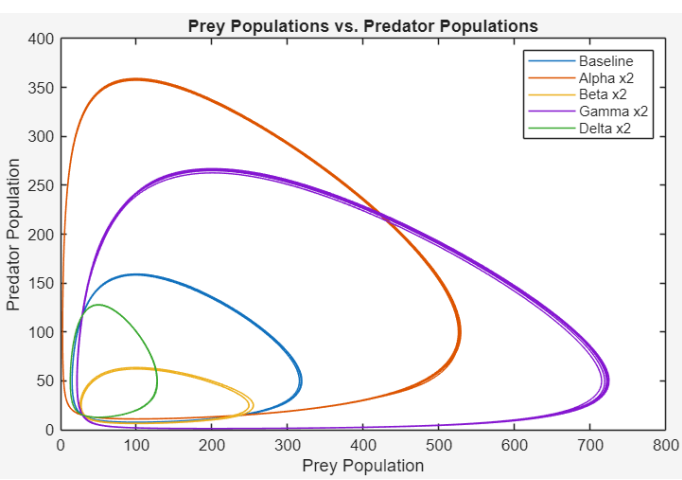


**FIGURE 3.25:** BASELINE, α×2, β×2, γ×2, AND δ×2 PREY POPULATIONS VERSUS PREDATOR POPULATIONS

## 4. RESULTS

Using the numerical model with the baseline parameters provided, the predator population reached a maximum of 158 individuals, while the prey population reached a maximum of 318 individuals. In addition, both populations exhibited periodic oscillatory behavior, with a peak to peak cycle of approximately 7.4 years. The minimum population values during the cycles were 8 individuals for the predator population and 15 individuals for the prey population.

## 5. DISCUSSION

### 5.1 Effect of Prey Growth Rate, α

Increasing $\alpha$ by a factor of 2 increased the prey growth rate, enabling the prey population to grow and recover more rapidly before the predator population could fully respond. As shown in Figures 3.23-3.25, this led to higher peak prey population. The increased prey availability subsequently supported greater predator growth, which also increased the peak predator population. Once the predator population became sufficiently large the increased predation pressure drove down the prey population to a lower minimum. The predator population was able to maintain a higher minimum than before from the earlier prey surplus and a shorter cycle period that resulted in less time spent in low prey conditions.

### 5.2 Effect of Predator Growth Rate, δ

When increasing δ by a factor of 2 the predator growth rate increased, leading to a more rapidly responding predator population. As a result, the prey population had less time to build up before the predator population responded leading to a decrease in both predator and prey peak populations. This also shortened the oscillation periods, likely due to the cycle progressing more rapidly with the quicker increase in predator populations. In this specific case the peak populations were equal between predator and prey, indicating the predator population limited prey growth before a substantial prey surplus could develop. This indicates that increasing δ strengthens the predator response and reduces the magnitude of the periods.

### 5.3 Effect of Predator Death Rate, γ

Increasing γ by a factor of 2 resulted in an increased predator death rate, which subsequentially led to decreased predation pressure due to a reduced predator population. As a result, the prey population was able to increase higher before the predator population recovered enough to suppress it. The larger peak prey population supported additional predator growth, leading to an increase in peak predator population as well. The oscillation period decreased slightly which could be partially due to the predator population decline more rapidly, allowing the prey population to recover sooner and start the next cycle earlier.

### 5.4 Effect of Predation Rate, β

Increasing β by a factor of 2 resulted in a higher predation rate, which reduced the ability of the prey population to build a surplus before being suppressed by the predators. As a result, both the prey and predator populations reached a lower peak value because of the increased predation pressure. In addition, this produced the lowest peak predator to prey ratio, with only approximately 0.25 predators per prey individual. This likely occurred because the higher predation rate suppresses the prey earlier which limits its ability to support predator growth for longer. Consequently, both species exhibited lower overall population levels. The oscillation period also decreased slightly by approximately 0.4 years, suggesting the stronger interaction between prey growth and predation pushed the system through the peaks faster.

## 6. CONCLUSION

This study aimed to conduct and analyze a sensitivity analysis on the variables α, β, γ, and δ within the Lotka-Volterra predator-prey model equations. The bassline case showed the periodic oscillatory behavior of both the predator and prey populations. Varying the parameters demonstrated that changes within the equation constants can significantly influence the maximum and minimum population levels of both species as well as changing the oscillation period length. These results demonstrate the high dependence of the Lotka-Volterra predator-prey model on its parameter constants, stressing the importance of accurate parameter selection when applying this model.

## 7. REFERENCES

[1] Volterra, Vito. “Variations and Fluctuations of the Number of Individuals in Animal Species Living Together,” 1928, pp. 22-23

[2] Herman, Edwin "Jed", and Gilbert Strang. Calculus Volume 1, 2016, pp. 426-429.